



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2022-23
HALF YEARLY EXAMINATION

CLASS X
SUBJECT: PHYSICS (SET B)

FULL MARKS: 80
TIME: 2 HOURS

Candidates are allowed additional 10 minutes for only reading the paper.

They must NOT start writing during this time.

The intended marks for questions or parts of questions are given in brackets []

This paper consists of seven printed pages

SECTION A

(Attempt all questions from this section)

[15]

- (i) The critical angle for glass is 42° . The corresponding angle of refraction r is
a) $r < 45^\circ$ b) $r = 90^\circ$ c) $45^\circ < r < 90^\circ$ d) $0^\circ < r < 45^\circ$
- (ii) Which of the following cannot be used as a force multiplier?
(a) single movable pulley
(b) single fixed pulley
(c) second order lever
(d) block and tackle system
- (iii) A machine has velocity ratio 4 and mechanical advantage nearly equal to 4. The friction in the machine increases for not taking proper care of it. Find the correct option
(a) Both velocity ratio and mechanical advantage will remain same
(b) Velocity ratio will decrease
(c) Mechanical advantage will increase
(d) Mechanical advantage will decrease
- (iv) For the same angle of incidence, angle of refraction in three media A, B and C are 15° , 25° and 35° . In which of the above media light travels with slowest speed?
(a) In B
(b) In C
(c) In A
(d) Unless the value of incident angle is known, it can't be determined
- (v) A concave lens of power P is immersed in water. The power will
(a) Increase (b) decrease (c) remains the same
(d) Increases for red color and decreases for blue color

(vi) 1. An object of length 5 cm is placed 25 cm away from a converging lens of focal length 10 cm. With respect to this, answer the questions that follow. The size of the image will be

- (a) +7.5 cm
- (b) -7.5 cm
- (c) +3.33 cm
- (d) -3.33 cm

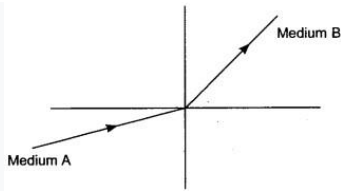
2. Position of the image will be

- (a) +16.7 cm
- (b) -16.7 cm
- (c) +7.14 cm
- (d) -7.14 cm

(vii) The energy of an electron is 4.8×10^{-19} J. When expressed in eV, the energy is

- (a) 1.6 eV
- (b) 3.2 eV
- (c) 3 eV
- (d) 6.5 eV

(viii) A light ray enters from medium A to medium B as shown in figure. The refractive index of medium B relative to A will be

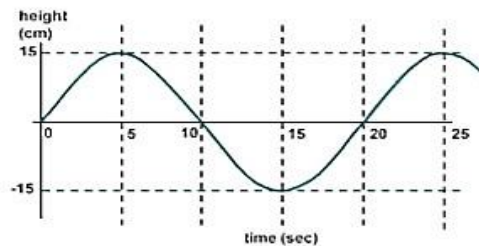


- (a) greater than unity
- (b) less than unity
- (c) equal to unity
- (d) zero

(ix) A hole is made in a convex lens. Then

- a) A hole appears in the image
- b) Image size decreases
- c) Image intensity decreases
- d) No change

(x) Observe the graph below and choose the option stating the correct values.

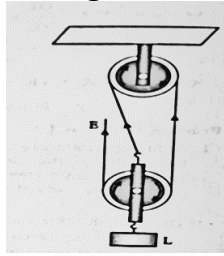


- (a) $T = 10$ s; Amplitude = 15 cm
- (b) Frequency = 0.05 Hz; Amplitude = 30 cm
- (c) $T = 20$ s; amplitude = 15 cm
- (d) Amplitude = 15 cm; Wave velocity = 20 m/s

(xi) Choose the correct option:

- (a) Infrasonic travels slowest of all
- (b) Ultrasonic travels faster than infrasonic
- (c) Audible sound travels faster than infrasonic
- (d) All travel at the same speed

(xii) The mechanical advantage of the following arrangement is

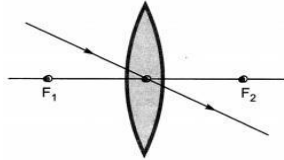


(a) 2

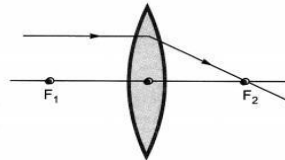
(b) 3

(c) 4

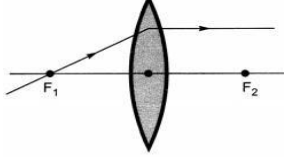
(xiii) The diagrams showing the correct path of the ray after passing through the



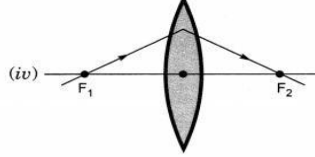
I



II



III



IV

(a) II and III only

(b) I and II only

(c) I, II and III

(d) II and IV

(xiv) Assertion- light ray passing through the optical centre of a thin lens travels undeviated. Reason- the central portion of a lens behaves like a very narrow glass block. Hence lateral displacement is negligible.

- (a) Both assertion and reason are correct and reason is the correct explanation of the assertion
- (b) Both assertion and reason are correct but reason is NOT the correct explanation of the assertion
- (c) Assertion is correct but reason is a wrong statement
- (d) Both assertion and reason are wrong statements

(xv) If a diverging lens having focal length f is to be used to form an image which is one fourth of the size of the object. Where must the object be placed?

(a) $3f$

(b) $4f$

(c) $2f$

(d) at F

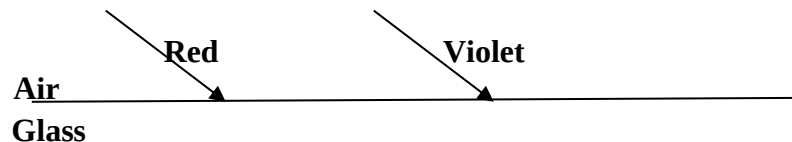
Question 2

- (i) (a) A block and tackle system has 5 pulleys. In ideal situation, how is the distance moved by the effort related to the distance moved by the load?
(a) For a given prism for white light, which color will have maximum deviation— red, blue or green?
(b) State the energy changes in a microphone. [3]
- (ii) (a) How does the frequency of a stretched string change, when tension is decreased by four times keeping other parameters constant?
(b) State one reasons why the efficiency of a single movable pulley is not 100%? [2]
- (iii) Can the moment of force be zero if the force is not zero? Justify your answer. [2]

- (iv) A postage stamp placed under a glass, appears to be raised by 8 mm. If refractive index of glass is 1.5, calculate the actual thickness of the glass slab [2]
- (v) The centre of gravity of two cones of same height is found to differ by 2 cm. When can this happen? Find the height. [2]
- (vi) State any two reasons that makes ultrasonic wave suitable for certain applications. [2]
- (vii) Draw a graph showing the variation of potential energy with height of a stone falling from a high tower and reaching the ground. [2]

Question 3

- (i) The speed of a wave in a medium is 960 ms^{-1} . If 3600 waves pass through a point in 1 minute, then calculate the wavelength of the wave. [2]
- (ii) A lens doesn't form image on screen. Identify the type of lens. Can this lens be used to produce magnified image? [2]
- (iii) When a troop crosses a suspended bridge, the soldiers are asked to break their steps. Give reason. [2]
- (iv) If in a machine the load moves at a greater speed than the effort, which one is greater – the load or the effort? Explain. [2]
- (v) Two parallel rays of red and violet color travelling through air, meet the air-glass boundary as shown in the diagram. Will their path inside the glass be parallel? Justify your answer. [2]

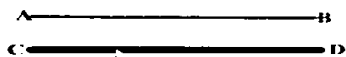


SECTION B

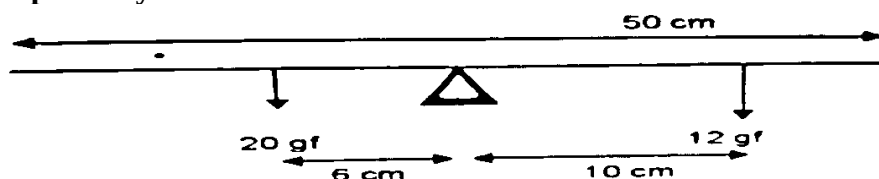
(Attempt any four questions)

Question 4

- (i) a) Define pitch
- b) Two wires AB and CD of same length are stretched by same amount. Which wire will produce sound of greater pitch on plucking? Give a reason for your answer. [3]



- (ii) A half metre long rod is pivoted at the centre, with two weights of 20 gf and 12 gf suspended at a perpendicular distance of 6 cm and 10 cm from the pivot point respectively as shown below.

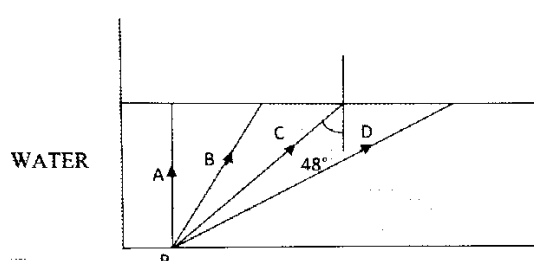


- a) Which of the two forces acting on the rigid rod causes anti-clockwise moment?
- b) Is the rod in equilibrium?
- c) The direction of 12 gf force is reversed. What is the magnitude of the resultant moment of the forces on the rod? [4]

- (iii) A block and tackle system of pulley has velocity ratio 3. Draw a labelled diagram of the system indicating clearly the points of application and direction of load and effort. [3]

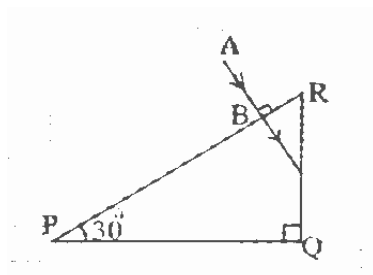
Question 5

- (i) A lens is used in a slide projector.
 a) Where is the object positioned?
 b) Where is the image formed by the lens?
 c) Draw a neat labelled diagram of the required arrangement. [3]
- (ii) A block and tackle pulley system having 3 pulleys and efficiency 80%. Find the effort required to lift a load of 150 N. [3]
- (iii) The diagram below shows a point source P inside a water container. Four rays A, B, C, D starting from the source P are shown upto the water surface. Draw the diagram and show the path of these rays after striking the water surface. The Critical Angle for water air surface is 48° . [4]



Question 6

- (i) A man standing in front of a wall produces a sound and hears an echo after 3 second. He walks 'x' m away from the wall and produces the same sound. Now he hears an echo after 5.25 second. Calculate the distance he walked away from the wall. [Speed of sound in air is 340 m/s]. [4]
- (ii) Redraw and complete the path of the ray AB till it emerges out of the prism of critical angle 42° . [3]

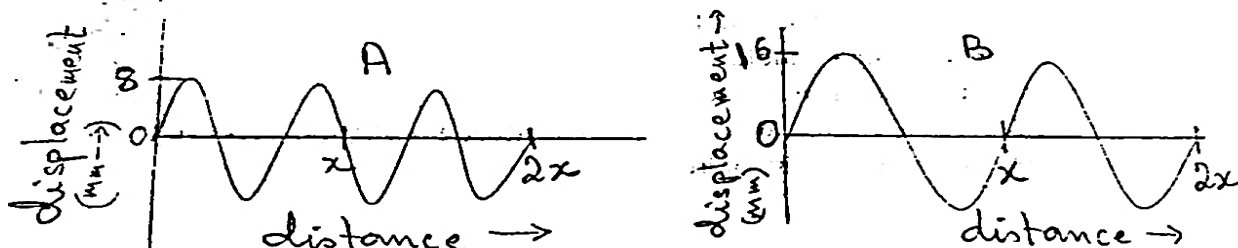


- (iii) a) Identify the wave produced by changes in the nucleus of an atom.
 b) Why is a rock salt prism used instead of a glass prism to obtain infra-red spectrum?
 c) Why are infra-red signals used during the war? [3]

Question 7

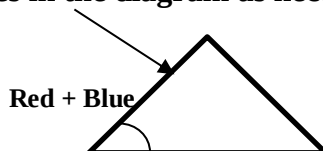
- (i) a) What are damped vibrations?
 b) Give one example of damped vibrations.
 c) Name the phenomenon that causes a loud sound when the stem of a vibrating tuning fork is kept pressed on the surface of a table.

- (ii) An object 4 cm high is placed at a distance of 40 cm from the optical centre of a concave lens of focal length of 20 cm. Calculate
- The position of the image.
 - The size of the image. [4]
- (iii) In the figure given below A and B represent the displacement-distance graph for two sound waves passing through air. What is the relation between their i) loudness and ii) pitch? [The graphs are drawn on same scale] [3]



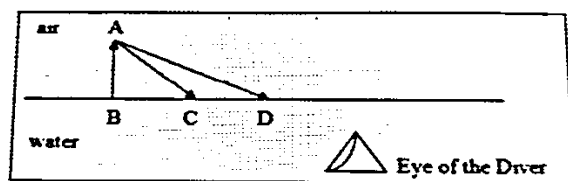
Question 8

- In a hydraulic power station, 1000 kg of water is allowed to drop through a height of 100 m in 1 s. If the conversion of potential energy to electric energy is 60%, calculate the power output. [$g = 10 \text{ ms}^{-2}$] [3]
- A ball of mass 100 g is thrown vertically upward with an initial velocity 20 ms^{-1} . Calculate the maximum height reached by the ball if 40% of the initial energy is lost due to friction of air. [Take $g = 10 \text{ ms}^{-2}$] [3]
- The diagram below shows a beam of light (red + blue) incident normally on an equilateral triangular prism. If the critical angle for the material of the prism is 60° for the light of red color, complete the diagram showing the path of each color emerging out of the prism. Show the angles in the diagram as necessary. [4]



Question 9

- A diver in water looks obliquely at an object AB in air.



- Does the object appear taller, shorter or of the same size to the diver?
- Show the path of two rays AC & AD starting from the tip of the object as it travels towards the diver in water and hence obtain the image of the object.
- Will there be any difference in the observation, if the positions of object and eye are reversed? [4]

- (ii) How does the angle of deviation formed by a prism change with the increase in the angle of incidence? Draw a graph showing the variation in the angle of deviation with the angle of incidence at a prism surface. [3]
- (iii) Mention the two conditions with the help of diagrams to alter the direction of rotation of an object. State principle of moment. [3]